Alex Moraru Thesis Companion

Exact Microscopic Reciprocity and Response Rigidity in Chiral Hexagonal Lattice Gases

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One paragraph abstract

The work asks when a chiral two dimensional transport response $Z = \bar{\eta} + i\eta_H$ is constrained to the circle anchored by its parity symmetric value and whether that constraint can survive beyond molecular chaos. Four results are separated. First, the deterministic HX89 endpoint automaton has an exact collision dressed reversor and an all time echo, supported by a complete 16,777,216 state certificate on the 2×2 six channel torus. Second, the published molecular chaos coefficients of the chiral FHP model lie exactly on a transport circle and satisfy a complex inverse shear eigenvalue relation [1]. Third, for any nonzero measured response, anchored circle membership is equivalent to constancy of $\mathrm{Re}(1/Z)$; C_6 symmetry gives the shear normal form $aI + bJ$ but does not force the dissipative coordinate a to be constant. Fourth, an exact stationary positive dissipation Ornstein Uhlenbeck Green Kubo family provides both rigid and nonrigid correlated classes, proving that symmetry and positivity alone do not protect the circle. The open problem is whether the fully correlated chiral FHP response obeys the inverse response rigidity condition under a specified finite size, time, frequency, or renormalization prescription.

Theorem, conjecture, and engineering separation

Statement	Status	Boundary
Collision dressed reversor and echo	Proved	Exact for the defined deterministic endpoint automata; does not determine viscosity.
Three one body collision invariants	Proved	Exact for the implemented local collision table.
Published chiral FHP kinetic circle	Exact corollary	Conditional on the published molecular chaos transport formulas.
$\Delta = 0 \iff \mathrm{Re}(1/Z) = \mathrm{Re}(1/Z_0)$	Proved	Closure free for every nonzero complex response with stated anchor.
C_6 implies $aI + bJ$	Proved	Symmetry normal form only; constant a does not follow.
Positive dissipation nonrigid family exists	Proved by construction	Exact OU Green Kubo counterexample; not an FHP surrogate.
Correlated chiral FHP preserves the circle	Open	Requires a microscopic, RG, resolvent, or equivalent correlated response proof.
Entropy sparse endpoint repair	Engineering	Seed 12345 and all ten signed exits terminate on explicit HALT cells; unrelated to the transport theorem.

Central equations

With $Z = x + iy$, $Z_0 = x_0 > 0$, define

$$\Delta = |Z|^2 - x_0 \operatorname{Re} Z. \tag{1}$$

Then

$$\frac{\Delta}{|Z|^2} = 1 - x_0 \operatorname{Re}\left(\frac{1}{Z}\right), \tag{2}$$

so

$$\Delta = 0 \quad \Longleftrightarrow \quad \operatorname{Re}\left(\frac{1}{Z}\right) = \operatorname{Re}\left(\frac{1}{Z_0}\right). \tag{3}$$

If $1/Z = a + ib$ and $a_0 = 1/x_0$,

$$\Delta = \frac{a_0 - a}{a_0(a^2 + b^2)}. \tag{4}$$

This is the complete membership classifier for the specified anchored circle.

For the published zero field chiral FHP molecular chaos formulas, define $D = \rho(1 - \rho/6)^3$, $d = p - 1/2$, $A = 1/(2D)$, and $u = (2\sqrt{3}/3)d$. Then

$$\bar{\eta} = \frac{A}{1 + u^2}, \qquad \eta_H = -\frac{Au}{1 + u^2}, \tag{5}$$

and therefore

$$\bar{\eta}^2 + \eta_H^2 = A\bar{\eta}. \tag{6}$$

The corresponding shear eigenvalue obeys $Z_B = -1/(4\lambda^*)$ in the repository convention.

Three strongest objections

1. The circle is only complex inversion

Complex inversion mapping a vertical line to a circle is classical. The work does not claim otherwise. The contribution is the exact anchored response classifier in terms of the measured inverse response, the symmetry boundary that shows C_6 does not force rigidity, and the explicit correlated witness on the nonrigid side. The limitation is that the general geometry of Mobius maps is prior mathematics, not a new result here.

2. Hexagonal symmetry should force the result

It does not. Solving the commutant of the two component shear representation gives $aI + bJ$. Reflection permits arbitrary even functions in $a(d)$ and odd functions in $b(d)$. The OU counterexample chooses a positive even drift $a = a_0 + \epsilon d^2$ and leaves the circle. A stronger microscopic principle could still protect a particular model, but it has not been proved for correlated chiral FHP.

3. The counterexample is not the FHP automaton

Correct. It is not used to predict the FHP outcome. It proves that stationarity, positive dissipation, reflection covariance, and the two component rotational form are insufficient by themselves. A model specific FHP proof therefore needs additional structure.

Why not alternatives?

Why not fit circles directly? A short noisy arc can fit many circles. The anchored inverse response criterion tests the exact relation and points directly to dissipative drift.

Why not use local curvature alone? A quartic dissipative drift has the rigid achiral curvature but leaves the circle away from the anchor.

Why not decide from the full collision spectrum? A Green Kubo response is a resolvent matrix element. Eigenvalues are insufficient without projectors and observable overlaps unless an exact reduced shear block and its normalization are identified.

Why not call the defect relation a Ward identity? No symmetry or conservation law has been shown to enforce constant $\mathrm{Re}(1/Z)$. Established viscosity Ward identities have a derivation from symmetry generators or conservation laws [2].

Glossary for a reader outside the subfield

Term	Meaning
Anchored response circle	The circle with diameter between the origin and the parity symmetric response value.
Circle defect	$\Delta = \lvert Z\rvert^2 - x_0 \,\mathrm{Re}\, Z$. It vanishes exactly on the anchored circle.
Green Kubo response	A transport response obtained from an integral of stationary time correlation functions.
Hall or odd viscosity	The parity odd reactive viscosity coordinate in a two dimensional fluid.
Inverse response rigidity	Constancy of $\mathrm{Re}(1/Z)$ relative to the achiral anchor.
Long time tail	Algebraic late time decay of a hydrodynamic correlation function. It affects limiting transport in two dimensions.
Molecular chaos	A kinetic closure that factorizes precollision correlations.
Shear doublet	The two real traceless symmetric shear components that rotate as a two dimensional representation.
Ward identity	A response identity derived from a specified symmetry or conservation law.

Prior art mention list

Reference	What it establishes and why it is needed here
Frisch, Hasslacher, Pomeau 1986 [3]	Establishes the triangular lattice gas route to Navier Stokes hydrodynamics. It supplies the foundational microscopic setting.
Ernst and Dufty 1989 [4]	Derives Green Kubo relations for lattice gas cellular automata from local conservation laws. It supplies the correlated transport language.
Naitoh, Ernst, Dufty 1990 [5]	Demonstrates long time correlation tails in two dimensional FHP fluids. It prevents an unqualified infinite time viscosity assumption.
Avron 1998 [6]	Gives the canonical formulation of odd viscosity as a nondissipative transport coefficient.
Banerjee *et al.* 2017 [7]	Develops odd viscosity in classical chiral active fluids and clarifies the reactive constitutive role.

Reference	What it establishes and why it is needed here
Bradlyn, Goldstein, Read 2012 [2]	Derives viscosity Kubo formulas and genuine Ward identities. It sets the standard for the thesis's Ward identity boundary.
Fruchart *et al.* 2022 [8]	Connects chiral collisions, non Hermitian kinetic operators, and odd transport. It motivates complex collision spectra without implying CRTC.
Allocca *et al.* 2026 [1]	Supplies the chiral FHP model and molecular chaos coefficients whose exact transport circle is reconstructed here.
Eren, Fruchart, Vitelli 2026 [9]	Develops a collisional kinetic model for odd fluids and uses collision spectra to identify hydrodynamic response structure.

Reproducibility manifest

The canonical machine tree and thesis are verified together. The primary command is:

```
make verify-all
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A successful run checks the HX89 machine suite, seed 12345 sparse termination, all ten entropy exits, sanitizer builds, the chiral certificate, 3,575 exact HX89 transport points, 35 CRTC tests, 9,595 exact zero field kinetic points, the 1,001 point counterexample domain, the independent symbolic thesis audit, generated fragment freshness, prose lint, PDF compilation, and unresolved reference checks.

The fully correlated chiral FHP circle remains an open problem after all of these checks. The verification suite establishes theorems and counterexamples under their stated definitions; it does not convert the open model specific statement into a theorem.

References

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[9] Ege Eren, Michel Fruchart, and Vincenzo Vitelli. Hydrodynamic description of collisional odd fluids from kinetic theory. *Physical Review E*, 114:024109, 2026. doi: 10.1103/v57s-653w.